Labour Market Integration of Immigrants in Canada Post-Pandemic (2021)

A Predictive and Descriptive Analysis

(Final Report)

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Research Questions & Contribution

This report addresses the following research questions:

1. To what extent can employment outcomes in Canada be explained by observable demographic and education-related characteristics?
2. Do age and education explain differences in employment outcomes across immigrant and non-immigrant groups?
3. Which modelling approach provides the strongest balance of effectiveness, efficiency, and stability for predicting employment status using the available features?

Main Contribution of the Study

This study contributes to the literature in three key ways.

First, it combines predictive modelling with descriptive labour market analysis. Rather than treating model performance as the sole outcome, the study uses predictive results to evaluate how well observable variables explain employment status, and complements this with subgroup comparisons across age and education categories.

Second, the study is situated in a post-pandemic labour market context. While the analysis is based on cross-sectional data from the 2021 Census of Population, it is interpreted in light of recent labour market shifts following COVID-19, including changes in employment patterns and immigrant integration outcomes (Hou, 2024).

Third, the study provides a structured model evaluation framework that compares classifiers based not only on effectiveness, but also on efficiency and stability. This allows for a more comprehensive assessment of model performance beyond predictive accuracy alone.

The core substantive contribution of the study is the finding that age and education are important correlates of employment outcomes, but they do not fully explain the persistent differences in employment rates between immigrant and non-immigrant groups. This suggests that additional structural and institutional factors play a meaningful role in shaping labour market integration.

Introduction

Labour market integration remains one of the most important dimensions of immigrant settlement in Canada. Immigration plays a central role in population growth, labour force sustainability, and long-term economic development. However, the transition into stable and equitable employment is not uniform across population groups. Despite Canada's points-based immigration system, which emphasizes education and labour market potential, differences in employment outcomes between immigrants and non-immigrants have historically persisted (OECD, 2019; Kaushik & Drolet, 2018).

Recent evidence suggests that immigrant labour market outcomes have improved since the mid-2010s, with employment rates rising and the overall gap between immigrants and the Canadian-born population narrowing (Hou, 2024). However, improvement does not necessarily imply full convergence. Differences in employment outcomes may persist even when immigrants possess similar levels of formal education or fall within comparable age groups as non-immigrants.

This study examines whether observable demographic and education-related characteristics, specifically age and education, can explain employment outcomes and whether differences between immigrant and non-immigrant groups remain after accounting for these factors. The analysis is based on aggregated data from the 2021 Census of Population, which provides detailed cross-tabulations of labour market outcomes across demographic and educational categories.

The study adopts a mixed analytical approach. The first component uses machine learning models to predict employment status from selected features. The second component uses descriptive subgroup analysis to compare employment rates across immigrant status, age groups, and education levels. Together, these approaches allow the study to evaluate both predictive performance and substantive labour market patterns.

The report is structured as follows. It first reviews relevant literature on immigrant labour market integration, then outlines the data, preprocessing steps, and modelling approach. It next evaluates model performance based on effectiveness, efficiency, and stability. The findings are then interpreted in the context of employment differences across demographic and educational groups. The final sections discuss limitations, ethical considerations, and directions for future research.

Literature Review & Integration

Labour Market Integration as a Multi-Dimensional Process

Labour market integration is widely understood as a multi-dimensional process that extends beyond formal human capital. While Canada's immigration system emphasizes education, skills, and labour market potential, research consistently shows that these characteristics do not automatically translate into equivalent employment outcomes.

The OECD (2019) highlights that immigrant qualifications and selection criteria do not guarantee successful labour market integration, even in a points-based system. Similarly, Kaushik and Drolet (2018) find that skilled immigrants in Canada continue to face barriers despite high levels of formal education. These findings help explain why strong educational attainment among immigrants may coexist with weaker employment outcomes or lower returns to education.

More recent Canadian evidence suggests that immigrant labour market outcomes have improved over time. Hou (2024) shows that employment rates among recent immigrants increased significantly between 2010 and 2023, narrowing the gap relative to the Canadian-born population. However, these improvements do not imply full convergence, as differences may persist within specific demographic or educational subgroups.

Credential Recognition, Work Experience, and Skill Transferability

One of the most important explanations for persistent employment differences is the incomplete transferability of foreign-acquired education and work experience. The Evaluation of the Foreign Credential Recognition Program (ESDC, 2025) identifies several key barriers faced by internationally trained individuals, including the time and cost of credential recognition, limited access to Canadian work experience, and challenges navigating professional licensing systems.

Kaushik and Drolet (2018) similarly emphasize institutional and structural barriers, including limited professional networks and difficulties accessing appropriate employment opportunities. These findings are particularly relevant to the present study. If employment differences persist even after accounting for age and education, it suggests that the remaining gap may be driven by institutional factors rather than differences in formal qualifications alone.

Language and Employment Outcomes

Language proficiency is another important factor linking immigrant characteristics to labour market outcomes. Xu and Hou (2023) find that official language proficiency is associated with improved labour market outcomes, particularly in terms of earnings. While its effect on initial employment may be more limited, language plays a significant role in long-term labour market integration and career progression.

This distinction is important for the current study. Even when age and education are observed, employment outcomes may still reflect underlying differences in language ability, social capital, and labour market signalling mechanisms.

Integrating the Literature with This Study

The present study builds on this literature by testing a focused question: whether observable demographic and education-related characteristics can explain employment outcomes and account for differences between immigrant and non-immigrant groups.

The findings provide only partial support for a human capital explanation. While employment rates vary across age and education categories, differences between immigrant and non-immigrant groups persist within comparable subgroups. This pattern is more consistent with literature emphasizing institutional barriers, skill transferability, and labour market frictions than with a model in which formal education and demographic characteristics fully determine employment outcomes (Kaushik & Drolet, 2018; OECD, 2019; Xu & Hou, 2023; ESDC, 2025).

In this sense, the study reinforces the view that labour market integration is not solely an individual-level outcome, but a process shaped by broader structural and institutional factors.

Methodology & Study Design

Data Source

This study uses aggregated data from **Statistics Canada Table 98-10-0442-01**, derived from the **2021 Census of Population**. The dataset provides cross-tabulated counts of labour force status by demographic and educational characteristics, including age group, gender, immigrant status, level of education, and field of study.

Unlike microdata sources such as the Labour Force Survey (LFS), this dataset is structured as an aggregated table. Each row represents a grouped combination of characteristics rather than an individual observation. As a result, the analysis is conducted at the group level, which has important implications for both modelling and interpretation.

The dataset contains a large number of observations due to the combination of multiple categorical variables across demographic and educational dimensions. This structure required preprocessing to isolate meaningful analytical units and remove non-informative aggregate categories.

Study Design

The project adopts a mixed-method approach:

1. A predictive modelling component, which frames employment status as a binary classification problem using aggregated observations.
2. A descriptive analytical component, which compares employment rates between immigrant and non-immigrant groups overall and within age and education categories.

This design allows the study to evaluate both the predictive power of observable variables and the persistence of group-level employment differences.

Data Cleaning and Preprocessing

Several preprocessing steps were applied to prepare the dataset for analysis:

- Rows representing aggregate totals or non-specific categories (e.g., “Total”) were removed.

- The dataset was restricted to observations classified as either *Employed* or *Unemployed*, excluding individuals outside the labour force.
- A binary target variable was constructed:
    - 1 = Employed
    - 0 = Unemployed
- Missing values were handled within the preprocessing pipeline.
- Categorical variables were transformed using one-hot encoding.
- Numeric variables were standardized where applicable.

These steps were necessary to convert the aggregated dataset into a format suitable for classification modelling and subgroup analysis.

Feature Selection

The predictive models used the following features:

- Geography
- Gender
- Age group
- Highest certificate, diploma, or degree
- Location of study
- Major field of study

These variables were selected because they represent key demographic and education-related characteristics that are plausibly associated with employment outcomes. They also align directly with the study's central research question regarding whether observable characteristics explain differences between immigrant and non-immigrant groups.

Model Selection

Two models were retained for the final analysis:

1. **Logistic Regression**
   This model serves as a transparent baseline and allows for interpretation of feature relationships through model coefficients.
2. **Random Forest**
   This model captures nonlinear relationships and interactions between variables and provides feature importance measures for comparison.

Gradient Boosting was initially explored but was excluded from the final workflow to maintain computational efficiency while still enabling meaningful model comparison.

Training, Testing, and Sampling Strategy

The dataset was divided into training and testing subsets using an 80/20 split with stratification on the target variable to preserve class balance.

Due to the size and structure of the dataset, stratified sampling was used to improve computational efficiency:

- Logistic Regression: 300,000 training rows
- Random Forest: 100,000 training rows

This approach is appropriate given the study's objective of comparative model evaluation rather than production-level optimization. The sampling strategy ensures reproducibility while maintaining sufficient data representation across categories.

Evaluation Metrics

Model performance was evaluated across three dimensions:

1. **Effectiveness**

   Measured using: Accuracy, Precision, Recall, F1 Score.
   The F1 score is emphasized as it balances precision and recall, providing a more informative measure when classification performance is weak or imbalanced.

2. **Efficiency**

   Measured using: Training time, Prediction time.
   These metrics assess whether increased model complexity leads to meaningful improvements in performance

3. **Stability**

   Stability was evaluated using 3-fold cross-validation on a stratified sample of 30,000 rows. This approach allows for assessment of model robustness across different data splits while maintaining computational feasibility.

Model Evaluation

Effectiveness

Model	Training Rows	Accuracy	Precision	Recall	F1 Score
Logistic Regression	300,000	0.4995	0.4995	0.5175	0.5083
Random Forest	100,000	0.4960	0.4964	0.5496	0.5216

Table 1: Model Performance Table

As shown in **Table 1** and illustrated in **Figure 1**, both models demonstrate weak predictive performance. Accuracy values remain close to 0.50, indicating that neither model performs substantially better than random classification. The F1 scores are also only marginally above 0.50, suggesting limited ability to balance precision and recall effectively.

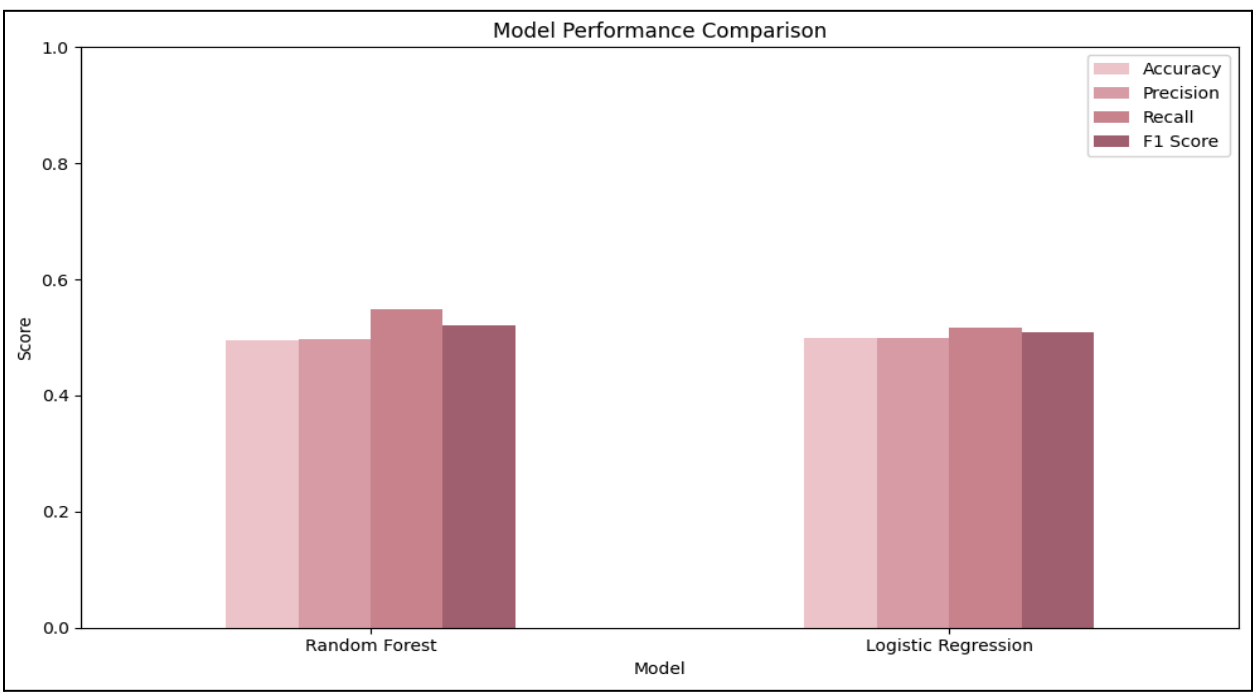

Figure 1: Model Performance Comparison

Among the two models, Random Forest achieves the highest F1 score (0.522), while Logistic Regression follows closely (0.508). However, as reflected in both the tabular results and the

visual comparison in Figure 1, the difference between the models is minimal and does not represent a meaningful improvement in predictive capability.

This result is substantively important. The consistently weak performance across both models suggests that the selected demographic and education-related features do not strongly determine employment outcomes within the dataset. In other words, while these variables are associated with employment, they are insufficient for accurate prediction on their own.

Efficiency

Model	Train Time (sec)	Test Time (sec)
Logistic Regression	0.60	0.44
Random Forest	0.48	0.61

Table 2: Efficiency Table

As presented in **Table 2** and visualized in **Figure 2**, both models demonstrate high computational efficiency under the sampled workflow. Logistic Regression has the fastest prediction time, while Random Forest shows slightly faster training time.

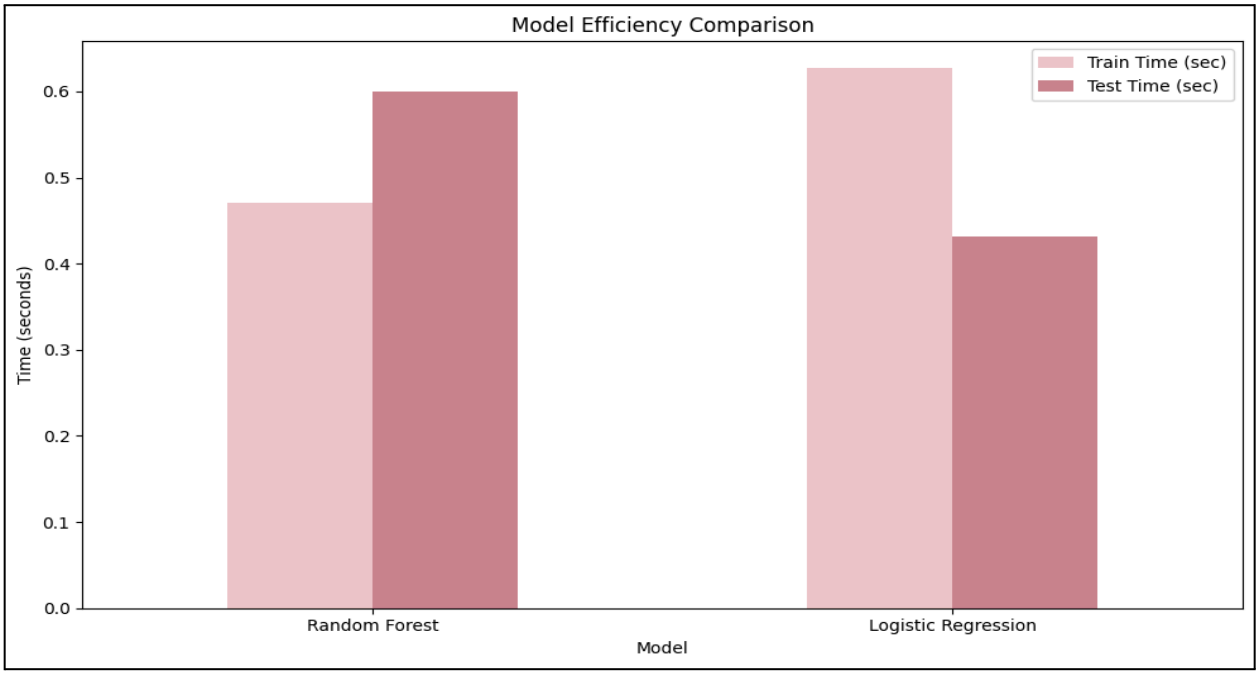

Figure 2: Model Efficiency Comparison

However, the differences are small and, as seen in Figure 2, do not meaningfully affect overall model comparison. Given the weak predictive performance observed in the effectiveness analysis, efficiency is not the deciding factor in model selection. Instead, the results suggest that increased model complexity does not lead to substantial gains in predictive accuracy within this feature set.

Stability

Model	CV Rows	Mean F1	Standard Deviation
Logistic Regression	30,000	0.4970	0.0034
Random Forest	30,000	0.4990	0.0158

Table 3: Cross-Validation Table

The cross-validation results shown in **Table 3** and summarized in **Figure 3** reinforce the findings from the test set. Both models produce mean F1 scores close to 0.50, indicating consistently weak predictive performance across different data splits.

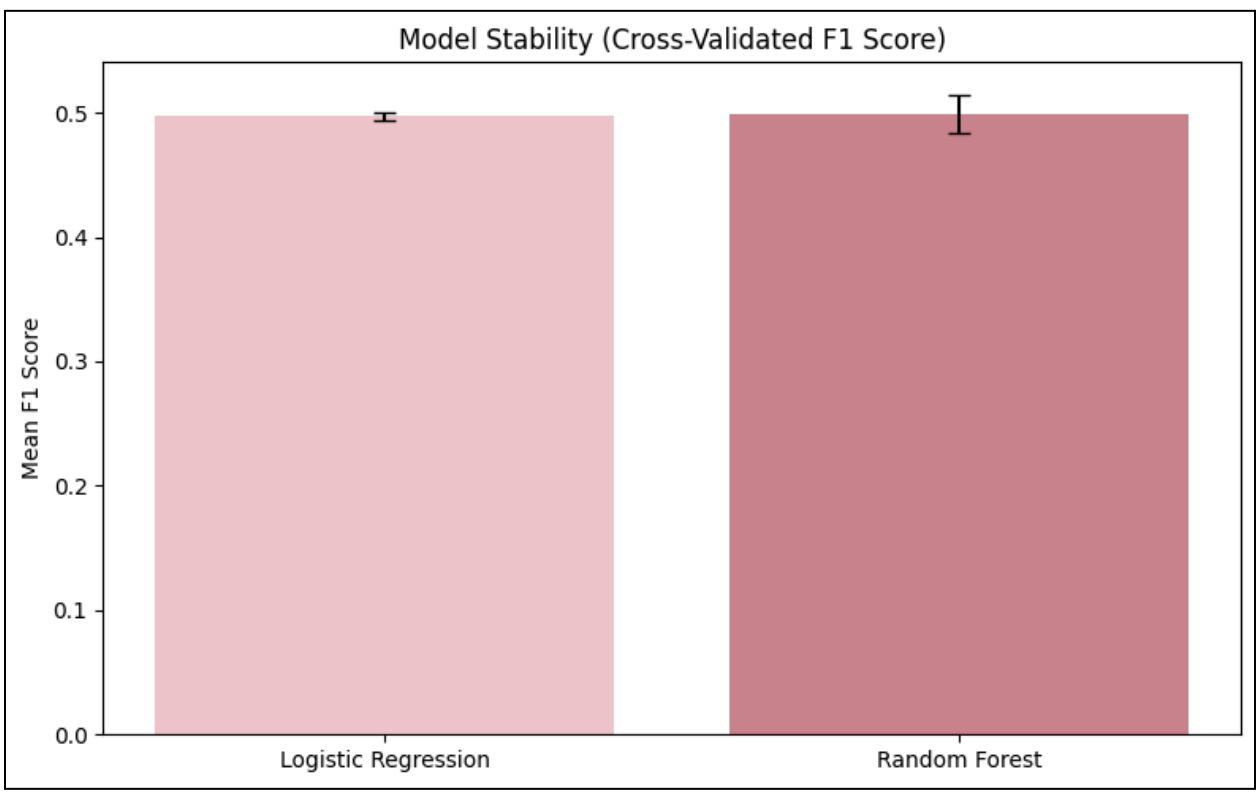

Figure 3: Cross-validated F1 comparison

Logistic Regression demonstrates lower variability across folds, suggesting more stable but still limited performance. Random Forest shows slightly higher average F1 scores but greater variability, as reflected in Figure 3, indicating less consistency across validation runs.

Overall, neither model demonstrates strong predictive stability in a substantive sense, as performance remains close to chance across all evaluation scenarios.

Model Selection

The final model choice depends on the evaluation criteria emphasized:

- For interpretability and consistency: Logistic Regression is preferable
- For marginally higher F1 performance: Random Forest performs slightly better

However, when considering effectiveness, efficiency, and stability together, the most important conclusion is that neither model achieves strong predictive performance.

This conclusion is supported consistently across **Tables 1–3** and **Figures 1–3**. Rather than indicating a limitation of the modelling approach, these results reflect the structure of the dataset and the scope of the selected features. Because the analysis is based on aggregated observations and a limited set of demographic and education-related variables, the models are constrained in their ability to capture the full set of factors influencing employment outcomes.

The inclusion of both a linear model (Logistic Regression) and a nonlinear model (Random Forest) was intended to test whether more flexible modelling approaches would yield meaningful improvements in performance. The results show that allowing for nonlinear relationships and interactions does not substantially improve predictive accuracy, suggesting that the limitation lies in the available features rather than in model complexity.

More complex models, such as Gradient Boosting, were considered but not retained in the final workflow. Given the minimal performance differences observed between the selected models, additional complexity would likely increase computational cost without providing meaningful gains in predictive performance.

Overall, the model comparison reinforces the central finding of the study: observable demographic and education-related variables do not strongly predict employment outcomes. This supports the interpretation that unobserved structural and institutional factors play a significant role in shaping labour market integration.

Findings & Interpretation

Overall Employment Comparison

The descriptive analysis shows that non-immigrants have a slightly higher overall employment rate than immigrants:

- Non-immigrant employment rate: **~92.8%**
- Immigrant employment rate: **~90.6%**

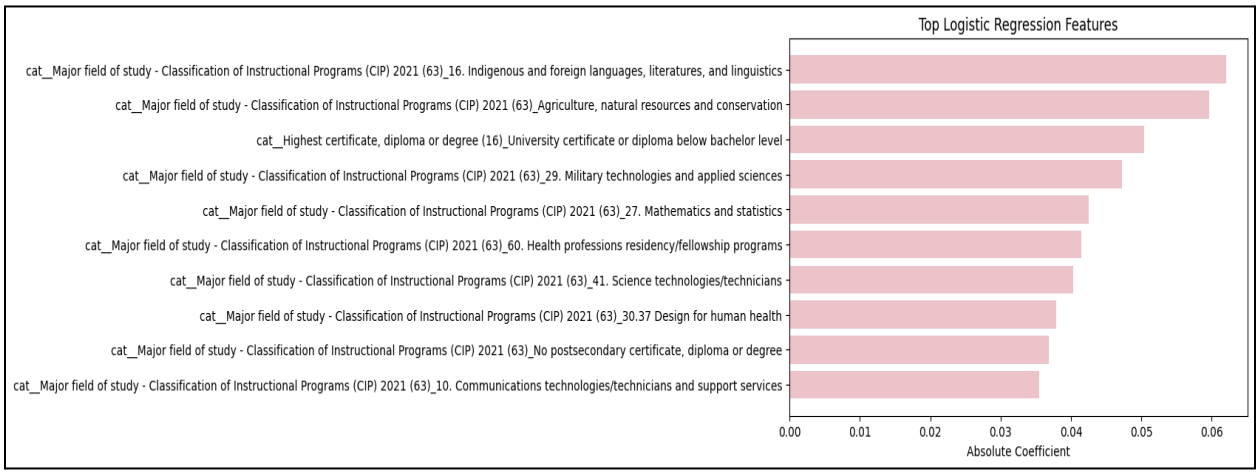

Figure 4: Top Logistic Regression features

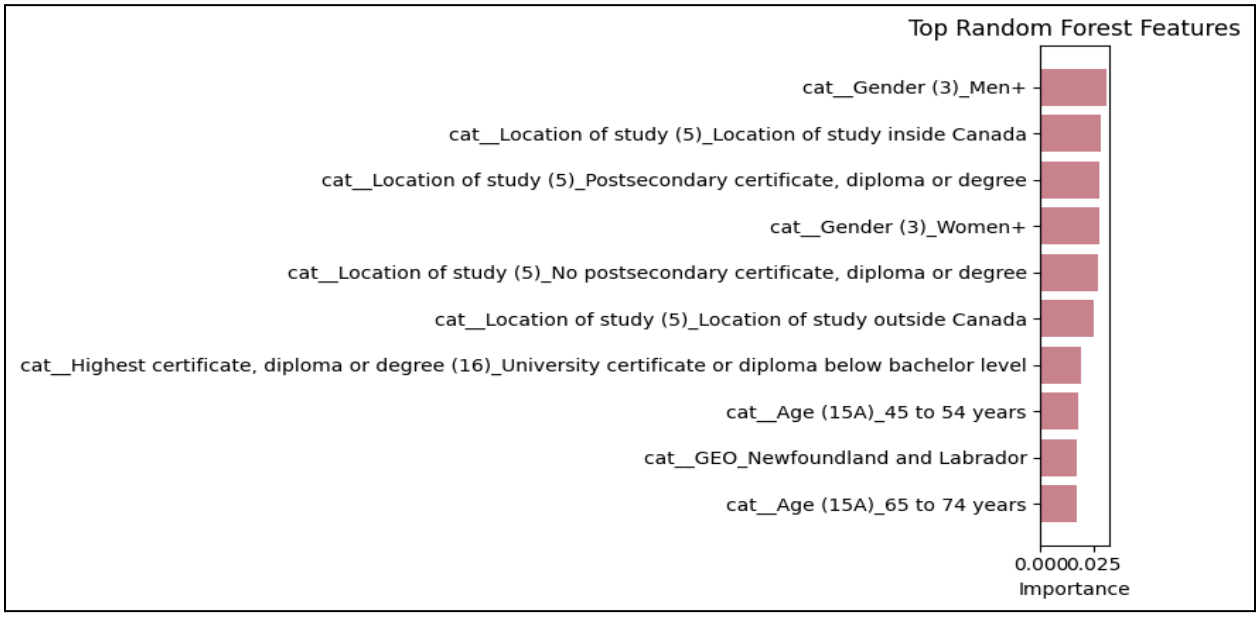

Figure 5: Top Random Forest features

Although the difference is modest, it is consistent. This establishes an important baseline pattern for the analysis. The key question is whether this gap disappears when comparing individuals with similar levels of education and age.

Feature importance results from both models (Figures 4 and 5) indicate that variables such as field of study, location of study, and age contribute to prediction. However, given the overall weak model performance, these results should be interpreted cautiously and used primarily to support broader descriptive findings rather than as definitive explanations.

Does Education Matter?

Education is clearly associated with employment outcomes, but it does not fully explain the difference between immigrant and non-immigrant groups.

As shown in **Figure 6**, employment rates are generally higher for individuals with stronger postsecondary qualifications across both groups. This aligns with established labour market expectations and with Canadian immigration policy, which prioritizes education and skill-based selection (OECD, 2019).

However, as illustrated in **Figure 7**, non-immigrants maintain slightly higher employment rates across most education categories. While the differences are not large, they are persistent rather than disappearing. This suggests that education alone is not sufficient to eliminate the employment gap.

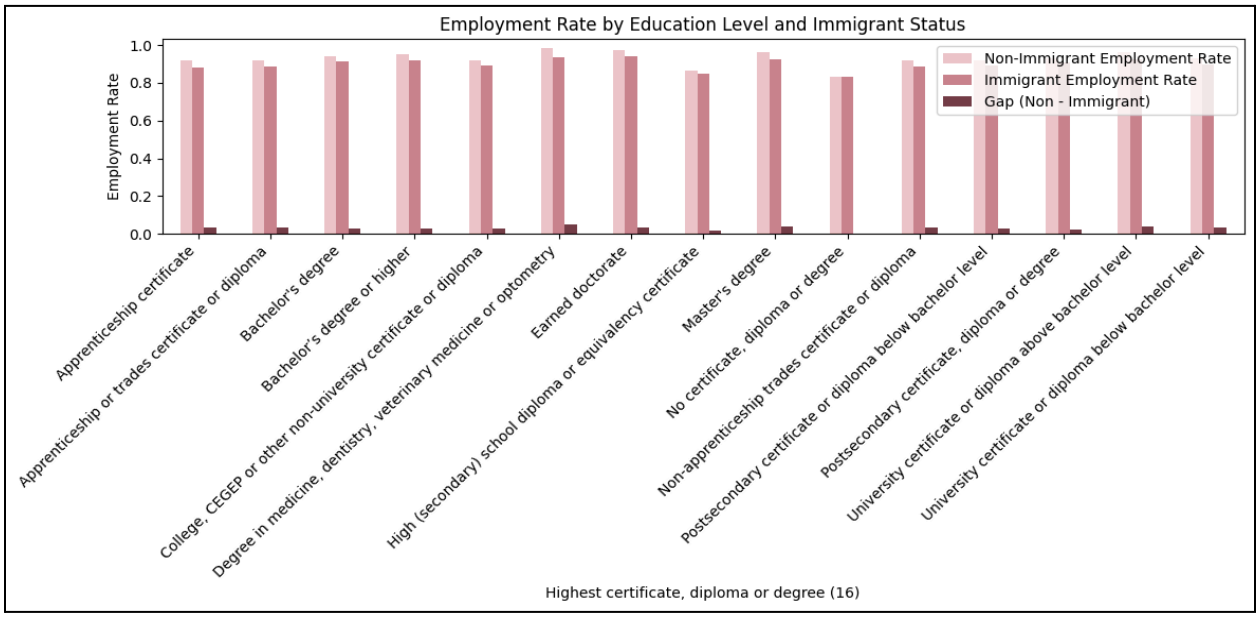

Figure 6: Employment rate by education level and immigrant status

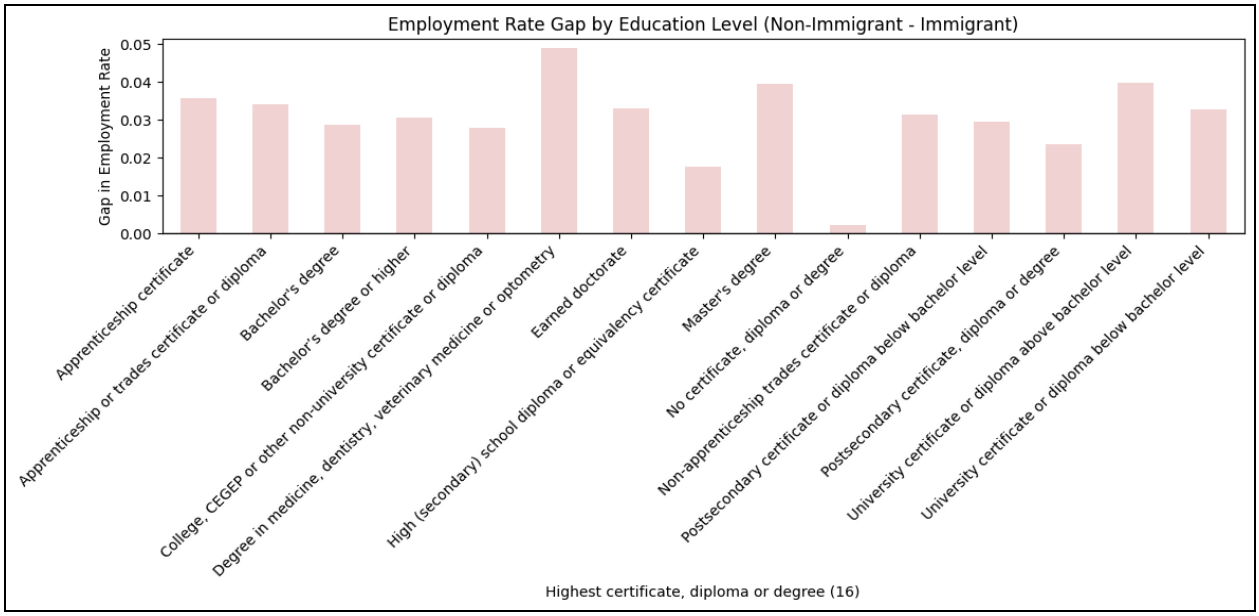

Figure 7: Employment rate gap by education level

This finding is consistent with prior research. The literature shows that the value of education depends on how it is recognized and utilized within the labour market. Factors such as location of study, credential recognition, and occupational requirements influence whether education translates into employment opportunities (Schimmele & Hou, 2024; ESDC, 2025).

In this context, the result can be summarized as follows:

- Education matters because employment rates vary across educational categories
- Education is not sufficient because group differences persist within those categories

Does Age Matter?

Age is also an important factor in employment outcomes, but it does not fully explain the observed group differences.

Employment rates vary across age groups, with lower attachment typically observed among younger individuals and those approaching retirement age. This pattern is consistent with expected labour market dynamics.

As shown in **Figure 8**, employment rates differ across age groups for both immigrants and non-immigrants. However, **Figure 9** shows that the gap between the two groups does not disappear within these categories. The differences tend to be more pronounced among younger age groups and smaller in some older categories, but they remain present overall.

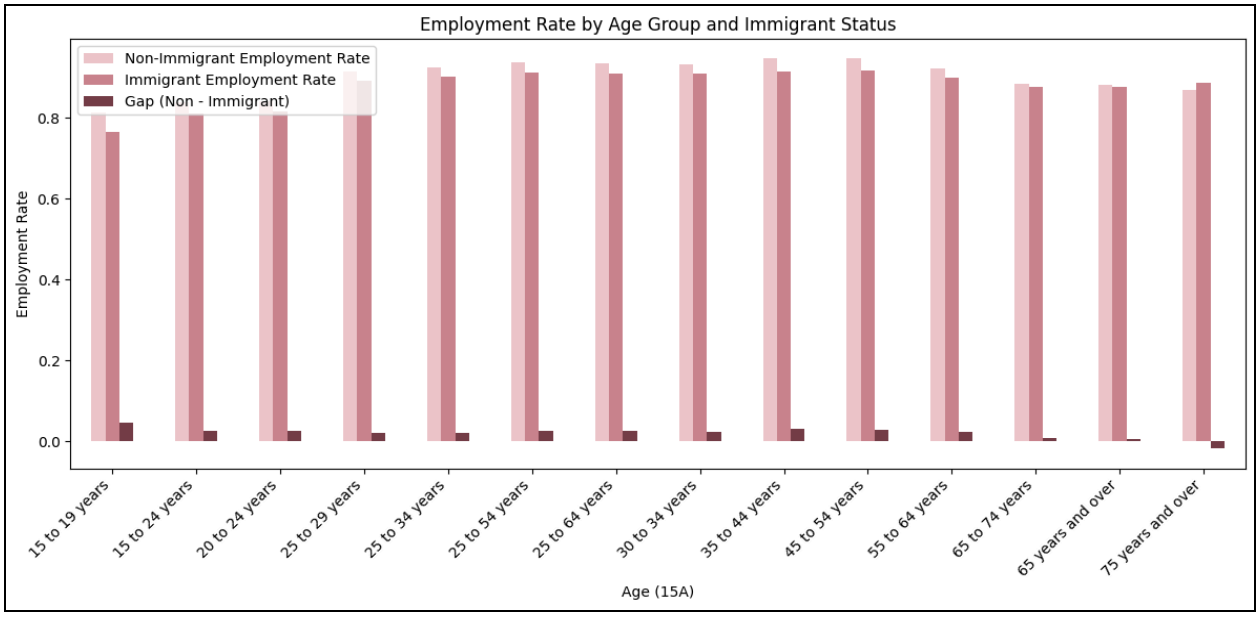

Figure 8: Employment rate by age group and immigrant status

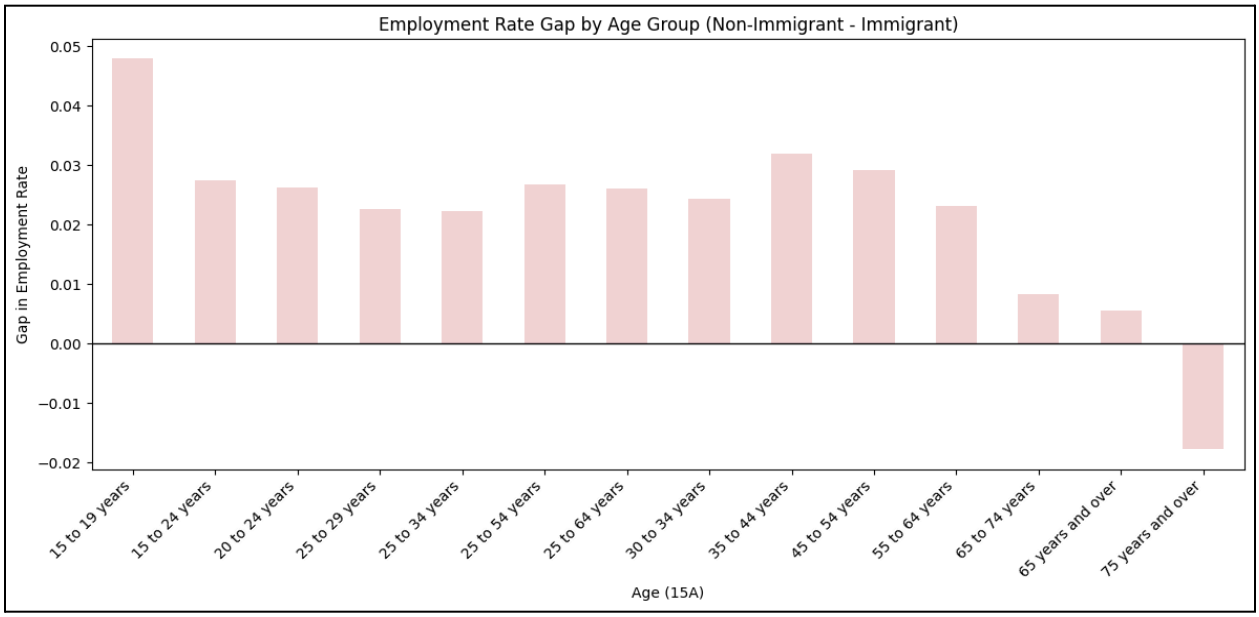

Figure 9: Employment rate gap by age group

This suggests that age plays a role in shaping employment outcomes but does not fully account for the persistent differences between groups.

One interpretation is that labour market entry may be particularly challenging for immigrants, especially younger individuals or recent entrants. This aligns with prior research highlighting barriers related to work experience, professional networks, and labour market entry pathways (Kaushik & Drolet, 2018; OECD, 2019).

Feature Interpretation

The feature importance analysis suggests that the models were most sensitive to variables related to:

- field of study
- location of study
- age
- geography
- gender

In Logistic Regression, several field-of-study categories show strong coefficients, while Random Forest places greater emphasis on variables such as gender, location of study, and age. However, these findings should be interpreted with caution.

Given that both models demonstrated weak predictive performance, the feature importance results do not provide strong causal or explanatory evidence. Instead, they indicate that while observable characteristics are relevant, they do not capture the full complexity of employment outcomes.

Additional Interpretations of Age and Education Patterns

A more detailed examination of subgroup patterns suggests that the relationship between immigrant status and employment outcomes may vary across both age and education in ways that reflect broader labour market dynamics.

Across age groups, the employment gap between immigrants and non-immigrants appears to be more pronounced among younger individuals and narrows in older age categories. One possible interpretation is that labour market entry may be more challenging for younger immigrants, particularly those who arrive as students or dependents. In these cases, limited Canadian work experience and language barriers may affect early employment outcomes.

At the same time, the narrowing gap among older age groups may reflect a gradual process of labour market integration. Over time, immigrants may accumulate local work experience, develop professional networks, and improve language proficiency, all of which can contribute to improved employment outcomes. However, this convergence does not necessarily imply equivalent long-term outcomes. Differences in job quality, income stability, access to benefits may still persist; potentially causing a lack of savings and the inability to retire.

Differences across education levels also reveal important patterns. The largest employment gaps appear in highly specialized fields, such as medicine, where international credentials may not be fully recognized or transferable to Canadian regulatory standards. This interpretation is consistent with existing research on credential recognition and occupational licensing barriers (ESDC, 2025).

In contrast, the smallest gaps are observed among individuals with little or no formal education. One possible explanation is that low-skilled occupations may rely less on formal credential recognition or be more accepting of language barriers, allowing immigrants and non-immigrants to compete more directly in these segments of the labour market. While this may reduce observable employment differences, it does not necessarily imply equivalent job quality or long-term economic outcomes.

These patterns reinforce the broader conclusion of the study: employment outcomes are shaped not only by observable characteristics such as age and education, but also by institutional factors that influence how skills and experience are recognized and rewarded within the labour market.

Synthesis of Findings

Taken together, the modelling and descriptive results support three main conclusions:

1. Age and education are important correlates of employment outcomes.
2. Observable demographic and education-related variables do not strongly predict employment status.
3. Differences between immigrant and non-immigrant groups persist even within comparable age and education categories.

This is a meaningful analytical result. It suggests that the remaining employment gap is likely influenced by factors not captured in the dataset, including language proficiency, credential recognition, labour market experience, and institutional barriers.

Discussion of Research Impact and Practical Implications

The findings have both methodological and policy implications.

From a methodological perspective, the study demonstrates the value of combining predictive modelling with descriptive subgroup analysis. Model performance alone would suggest weak predictive power, but when paired with subgroup comparisons, a more informative conclusion

emerges: observable characteristics are associated with employment outcomes but do not fully explain group differences.

From a policy perspective, the results suggest that improvements in formal education alone are unlikely to eliminate employment gaps. While immigrant labour market outcomes have improved in recent years (Hou, 2024), persistent differences indicate that structural barriers remain.

This has practical implications in several areas:

1. **Credential recognition:** Employment gaps among highly educated individuals suggest that the issue is not education itself, but how it is evaluated and utilized within the labour market.
2. **Early labour market integration:** Stronger differences among younger groups highlight the importance of transition supports, including internships, co-op opportunities, and employer connections.
3. **Settlement and language support:** Even though language is not directly modelled, prior research shows it remains an important determinant of labour market outcomes (Xu & Hou, 2023).

Overall, the findings support an interpretation of labour market integration as an institutional and structural process, rather than one determined solely by individual characteristics.

Limitations & Ethical Considerations

Limitations

This study has several important limitations that should be considered when interpreting the results.

First, the dataset is **aggregated** rather than individual-level. Each observation represents a grouped combination of characteristics rather than a specific individual. This limits the ability to capture within-group variation and may reduce the precision of predictive modelling. As a result, the models are constrained in their ability to identify nuanced relationships between variables and employment outcomes.

Second, the feature set is limited to observable demographic and education-related variables. Important factors such as years since immigration, language proficiency, occupational licensing, Canadian work experience, and social networks are not directly captured in the dataset. Prior research suggests that these variables play a significant role in shaping immigrant labour market outcomes (Kaushik & Drolet, 2018; Xu & Hou, 2023; ESDC, 2025). Their omission likely contributes to the weak predictive performance observed in the modelling results.

Third, the modelling approach relies on **stratified sampling** to ensure computational feasibility. While this is appropriate for comparative model evaluation, it means the models are not optimized for maximum predictive performance across the full dataset. Instead, the focus is on identifying patterns and limitations within a manageable and reproducible framework.

Fourth, the analysis is **cross-sectional**, based on data from the 2021 Census of Population. This limits the ability to examine changes over time or establish causal relationships. Although the study is interpreted within a post-pandemic context, it does not directly capture temporal dynamics in labour market outcomes.

Finally, an important limitation is the use of aggregated Census data rather than **individual-level microdata** such as the Labour Force Survey (LFS). While Census data provides broad coverage and detailed subgroup breakdowns, LFS data would allow for more granular modelling, including individual-level variation and time-series analysis. This distinction is important because it highlights the trade-off between data coverage and analytical depth. These limitations are not simply constraints but also help explain the study's findings, particularly the weak predictive performance observed across models.

Ethical Considerations

Several ethical considerations arise in the use and interpretation of labour market data.

First, there is a risk of **bias and misinterpretation** when using demographic variables such as immigrant status, age, and gender. These variables are socially meaningful and may reflect structural inequalities rather than individual capabilities. As a result, model outputs should not be interpreted as evidence of inherent differences in employability between groups.

Second, the **weak predictive performance** of the models has important ethical implications. Models that perform only slightly better than random classification should not be used in real-world decision-making contexts, such as hiring or screening. The findings of this study are intended for analytical and research purposes only and should not be applied at the individual level.

Third, there is a risk of **overgeneralization** when interpreting aggregate results. The findings describe group-level patterns and do not reflect individual outcomes. Care must be taken to avoid reinforcing stereotypes or attributing differences in employment outcomes to individual deficiencies rather than broader structural and institutional factors.

Finally, the study relies on official statistical data, which requires responsible interpretation. While the results identify persistent differences in employment outcomes, they should be understood within the broader context of labour market integration, including barriers related to credential recognition, labour market entry, and institutional processes (Schimmele & Hou, 2024; ESDC, 2025).

Future Work & Recommendations

Future Research

Several extensions of this study could deepen the analysis and address its current limitations.

First, future research could incorporate multiple Census periods, including earlier cycles such as 2017 or older and newer releases such as the 2025 Census when available. A multi-period approach would allow for a more robust examination of long-term trends in immigrant labour market integration, including whether observed employment gaps are stable over time or changing. This would also make it possible to assess whether the COVID-19 period had a lasting impact on employment outcomes or whether observed patterns reflect longer-standing structural differences.

Second, future work could integrate individual-level data from the **Labour Force Survey (LFS)** to complement the aggregated Census analysis. While the Census provides strong coverage and subgroup detail, LFS data would allow for more granular modelling and the inclusion of additional variables such as job tenure, occupation, industry, and hours worked. Combining these sources would enable a more comprehensive analysis of both employment status and the underlying mechanisms driving labour market outcomes.

Third, future research could explore differences in occupational and industry distributions across immigrant and non-immigrant groups. The findings of this study suggest that employment gaps persist even when age and education are similar. One possible explanation is that individuals may be concentrated in different sectors of the labour market, with varying levels of job stability, income, and advancement opportunities. Examining industry and occupation data could help determine whether these structural differences contribute to observed employment patterns.

Fourth, future work could investigate labour market entry pathways, particularly for younger immigrants. The current findings suggest that employment gaps are more pronounced among younger age groups. This raises important questions about the role of factors such as student status, early career transitions, language proficiency, and access to Canadian work experience. A more detailed analysis could help identify whether these factors contribute to initial employment disadvantages and how they evolve over time.

Finally, future research could incorporate additional variables related to credential recognition and skill transferability, particularly in highly regulated fields such as healthcare and professional services. The present study suggests that employment gaps are larger in certain education categories, which may reflect barriers to the recognition of international qualifications. Further investigation could help quantify the extent to which these barriers affect employment outcomes across different sectors. Together, these extensions would move the analysis from identifying patterns to more fully explaining the mechanisms underlying immigrant labour market outcomes.

Practical Recommendations

The findings of this study suggest several practical implications for labour market policy and integration strategies.

1. **Strengthen credential recognition systems**
   Persistent employment gaps among highly educated individuals indicate that the issue is not solely educational attainment, but how credentials are evaluated and applied within the labour market. Improving the efficiency and transparency of credential recognition processes could help reduce these barriers.
2. **Support early labour market integration**
   The larger employment gaps observed among younger individuals suggest that labour market entry may be particularly challenging for immigrants. Expanding access to internships, co-op programs, and entry-level opportunities could help facilitate smoother transitions into the workforce.
3. **Enhance language and workplace readiness programs**
   Although language proficiency is not directly measured in this study, prior research indicates its importance. Continued investment in language training and workplace communication support may improve employment outcomes, particularly in early career stages.
4. **Improve access to Canadian work experience**
   Many employment barriers are linked to the requirement for local experience. Creating more structured pathways—such as mentorship programs, bridging programs, and employer partnerships—could help immigrants gain relevant experience more quickly.
5. **Encourage equitable evaluation of international education and experience**
   Employers play a key role in labour market integration. Encouraging more consistent and informed evaluation of international qualifications could help reduce disparities in employment outcomes.

Code & Reproducibility

This project includes a reproducible analytical workflow implemented in Python for data preprocessing, modelling, evaluation, and visualization. The analysis is based on aggregated data from **Statistics Canada Table 98-10-0442-01**, derived from the **2021 Census of Population**.

The reproducible workflow includes:

- data cleaning and preprocessing steps applied to the aggregated table
- a Jupyter Notebook used for model training, testing, and subgroup analysis
- exported result files summarizing model performance, cross-validation results, and feature importance
- visual outputs used to support the figures presented in the report

Repository Requirements

https://github.com/jayemm2210/TMU_CIND820

Suggested Reproducibility Statement

This project was designed with reproducibility in mind. All data preprocessing, modelling, evaluation, and visualization steps are fully documented in the accompanying Python notebook and repository files. The analysis is based on aggregated data from Statistics Canada Table 98-10-0442-01, derived from the 2021 Census of Population.

To ensure computational feasibility, stratified sampling was used during model training and cross-validation. These decisions are explicitly documented in the workflow and were applied consistently across all models to maintain comparability.

Because the dataset is aggregated rather than individual-level, the results reflect group-level patterns rather than individual predictions. Within these constraints, the provided code and documentation are sufficient to reproduce the analytical process, model comparisons, and descriptive findings presented in this study.

Conclusion

This study examined whether employment outcomes in Canada can be explained by observable demographic and education-related characteristics, and whether differences between immigrant and non-immigrant groups persist when age and education are taken into account.

Using aggregated data from the 2021 Census of Population, the analysis combined predictive modelling with descriptive subgroup comparisons. The modelling results showed that both Logistic Regression and Random Forest performed weakly, with accuracy close to 50% and F1 scores only slightly above 0.50. These results indicate that the selected variables, primarily age and education, do not strongly predict employment status.

The descriptive analysis provides the study's most important substantive findings. Both age and education are clearly associated with employment outcomes. However, they do not fully explain the observed differences between immigrant and non-immigrant groups. While employment rates generally improve with higher education and vary across age groups, a consistent gap remains across most comparable categories.

Further examination of subgroup patterns suggests that this gap is not uniform. Differences are more pronounced among younger individuals and tend to narrow in older age groups, which may reflect gradual labour market integration over time. Similarly, variation across education levels indicates that employment gaps are larger in highly regulated or specialized fields, where credential recognition may play a role, and smaller in lower-skilled categories where formal qualifications are less central.

Taken together, these findings suggest that employment outcomes are shaped not only by observable characteristics, but also by structural and institutional factors that influence how skills and experience are recognized within the labour market. This interpretation is consistent with existing research emphasizing the importance of credential recognition, language proficiency, and access to Canadian work experience.

Overall, the study demonstrates that while demographic and education-related variables are important, they are not sufficient to explain employment outcomes or eliminate group-level differences. Future research should therefore focus on incorporating more detailed individual-level data and exploring the mechanisms underlying labour market integration, particularly in relation to institutional barriers and labour market entry pathways.

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